

# Beamer<sup>+</sup>

## Interactive Presentations

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# The Team

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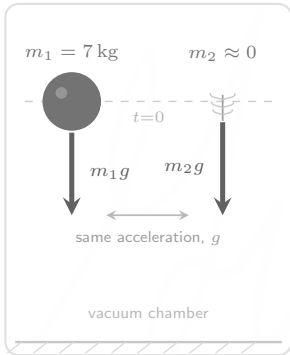
Chenhao Zhou

Beamer<sup>+</sup> is a lightweight web app that enhances static slides with embedded media, real-time drawing tools, intuitive navigation, and a suite of powerful presentation features.

## videos/audio

supported formats include .mp4, .mov, .mpeg, .mp3, .wav

example usage: demonstrating Newton's 2<sup>nd</sup> law in a physics course



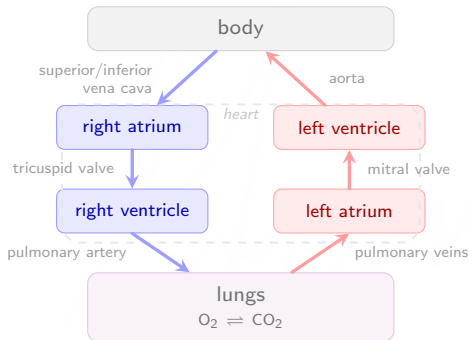
gravitational acceleration is  
independent of mass

 try clicking!

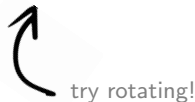
# models

supported formats include .glp**b**

example usage: explaining blood flow through the heart in an anatomy course



blood flow through the heart  
(deoxygenated blood / oxygenated blood)



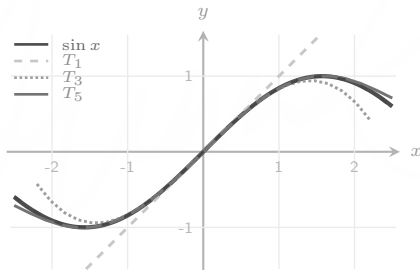
# widgets

supported formats include .html

example usage: explaining Taylor series in a calculus class

Any smooth function  $f$  can be approximated around  $x_0 = 0$  as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$



expansion of  $f(x) = \sin(x)$  around  $x_0 = 0$



## **widget library**

a collection of commonly used tools

examples include: calculator, timer, and camera)

(calculator widget)

(timer widget)

(camera widget)

## **use in computer science**

a collection of commonly used tools  
(e.g., calculator, timer, webcam, etc.)

(survey widget)

## example 1

presenting an algorithm and then coding it live in a lecture  
deleting from a linked-list

Given a singly-linked list, delete a node at a specified position.



```
class Node:
    def __init__(self, data: int):
        self.data = data
        self.next = None

    def delete_node(head: Node, target: int) -> Node:
        pass
```

(technical widget)

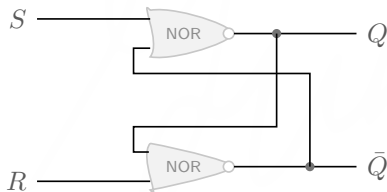
(ipy nb widget)

(quiz widget)

## example 2

demonstrating a technical concept with a virtual lab  
implementing an SR latch using NOR gates

A **latch** stores one bit of state using cross-coupled gate feedback.



cross-coupled NOR gates  
inputs are *active-high*

| $S$ | $R$ | $Q$ | $\bar{Q}$ | state     |
|-----|-----|-----|-----------|-----------|
| 0   | 0   | $Q$ | $\bar{Q}$ | hold      |
| 1   | 0   | 1   | 0         | set       |
| 0   | 1   | 0   | 1         | reset     |
| 1   | 1   | 0   | 0         | forbidden |

truth table for an SR latch  
implemented with NOR gates

(circuit widget)



(shell widget)

# using Beamer<sup>+</sup>

the steps to create a Beamer<sup>+</sup> presentation

(1) **create**

create slides using any standard tool (including L<sup>A</sup>T<sub>E</sub>X)

(2) **configure**

write `config` files to specify the placement and behaviour of any embedded media

(3) **package**

combine the slides, embedded media, and `config` files into a package

(4) **present**

open the package with Beamer<sup>+</sup>, which will display the slides with all embedded media integrated seamlessly

# creating a widget

a self-contained .html file that reads from a config object

widgets/hello.html

```
<!doctype html>
<html>
<body style="
  font-family: 'DM Sans', sans-serif;
  display: grid;
  place-items: center;
  height: 100vh;
  margin: 0;">

  <h1 id="msg"></h1>

  <script>
    // window.config injected by Beamer+
    const { name } = window.config;
    document.getElementById('msg')
      .textContent = 'Hello, ${name}!';
  </script>

</body>
</html>
```

config/s5.json

```
{
  "type" : "hello",
  "name"  : "Chandra"
}
```



Beamer<sup>+</sup> reads the config file and injects it before your script runs.

# package structure

one folder per presentation

```
my-lecture/  
|  
+-- slides.pdf  
|  
+-- audio/  
|   |-- lecture-intro.mp3  
|  
+-- video/  
|   |-- demo.mp4  
|  
+-- widgets/  
|   +-- hello.html  
|   |-- sr-latch.html  
|  
\-- config/  
    +-- s3.json  
    +-- s7.json  
    |-- assets/  
        |-- diagram.png
```

slides.pdf

your slide-deck as a PDF

audio/

narration, sound effects

video/

pre-recorded video clips

widgets/

one .html per widget

config/

one .json per slide

config/assets/

images or data files

config files are named `sNN.json` where `NN` matches the slide number in the PDF (slides with no media have no config file).

# widget prompt

a prompt for generative AI to produce widgets

prompt

```
Create an interactive HTML widget that <brief description>.
```

```
Output Ttwo files:
```

```
  widget.html  -- the widget itself
  config.json  -- an example config file with any relevant parameters
```

```
Widget rules:
```

- Single self-contained .html file  
(CDN links are fine, but no local imports)
- Read all parameters from window.config  
(this will be injected by another tool before the script runs)

```
Config rules:
```

- Valid JSON file
- Must contain the following: "type": <name>, "x": 0, "y": 0, "width": 1, "height": 1
- Any additional fields needed for the widget

```
Detailed Widget brief:
```

```
<describe in detail what you want here>
```

**learn more**

visit the link below to learn more

`chandra-gummaluru.github.io/post/beamer-plus`